PMVC-WNM: Phonetic Multi-View Co-training with Weakly Supervised Noise Modeling for Robust Banglish Sentiment Classification

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***Abstract*—Banglish (Bangla written in Roman script) is the dominant form of Bangladeshi social-media text, yet its lack of a fixed spelling convention leaves sentiment classification systems poorly served, particularly under the low-label-budget conditions typical of real annotation projects. We benchmark and diagnose a two-view co-training pipeline for this setting: View A uses character n-gram TF-IDF over orthographic surface forms, View B uses a phonetic (sound-normalized) encoding, and both are coupled with a weakly supervised noise model (WNM) that injects learned spelling-variant noise and gates pseudo-labels by cross-view agreement; we evaluate on a merged, class-balanced 11,788-row three-class dataset combining BnSentMix, b-and-b-80K, and EnBn-100K. Our reproduction of the deployed pipeline matches its own logged run to within 0.003 macro-F1, establishing the harness as a faithful basis for diagnosis. Decomposing the co-training gain shows that two-view ensembling alone, with no iterative pseudo-labeling, accounts for $+0.015$ macro-F1 over the single-view baseline, while the co-training loop itself contributes only a further $\sim +0.01$: most of the benefit comes from the ensemble, not from iteration. An exhaustive comparison of 21 classical classifiers across 7 families finds that `CalibratedClassifierCV(RidgeClassifier)`, previously untested in this setting, outperforms the deployed classifier on both views at the deployed 600-label budget ($+0.0118$ View A, $+0.0020$ View B), with the advantage holding at every tested label budget on View A and at budgets of 600 and above on View B, and under up to 35% injected spelling noise on both views. We further show that the deployed agreement gate's purity advantage over plain confidence filtering is front-loaded, winning at iteration 0 (0.929 vs. 0.916) but inverting by the final iteration (0.854 vs. 0.891; seed 42), a measured co-training degradation mode driven by declining view independence as pseudo-labels accumulate. Overall, PMVC-WNM does not significantly outperform standard co-training on this data and scale ($+0.0027$ macro-F1, inside a seed standard deviation of ≈ 0.01), and we treat that as the central result rather than a caveat: this paper's contribution is the diagnosis of where the gain comes from, not a claim that the gain itself is large.**

***Index Terms*—Banglish sentiment analysis, code-mixed text, co-training, semi-supervised learning, phonetic encoding, character n-grams, weakly supervised noise modeling, low-resource NLP, classifier calibration, pseudo-labeling**

I. INTRODUCTION

Code-mixed, Roman-script social media text, commonly termed *Banglish* and formed by transliterating Bangla into Latin characters and freely interleaving it with English, is ubiquitous across South Asian social platforms. It is nonetheless poorly served by standard NLP tooling trained on formal, monolingual corpora. The core obstacle is orthographic: Banglish has no fixed spelling convention, so a single word may surface in a dozen visually and phonetically related but string-dissimilar forms, undermining tokenizers, embeddings, and classifiers that assume orthographic regularity.

We focus this problem on sentiment classification under a low-label-budget regime: the realistic deployment scenario in which thousands of unlabeled comments are readily scraped but only a small fraction can be manually annotated. We instantiate this concretely with 600 labeled examples against a pool of roughly 9,430 available training rows, a labeling rate of approximately 6.4%, reflecting the annotation budgets typical of small research teams and low-resource languages rather than the fully-labeled corpora assumed by most benchmark work.

Prior Banglish sentiment classification work has largely pursued one of two directions: data augmentation to synthetically expand labeled sets [1, 2], or fine-tuning deep transformer models on whatever labeled data is available [3]. Neither line evaluates classical two-view co-training in the original sense of Blum and Mitchell [4], which trains two conditionally-independent-view classifiers that iteratively pseudo-label each other's confident predictions, as a low-resource remedy for this specific language setting. Nor does any prior study rigorously diagnose why co-training helps when it does: gains in the literature are typically presented as a single aggregate improvement, without decomposing how much of that gain comes from ensembling two views at all versus from the iterative pseudo-labeling process itself. This paper makes that distinction explicit.

This paper makes the following contributions:

- We reproduce and rigorously benchmark a two-view (orthographic + phonetic) co-training pipeline for Banglish sentiment classification, validating our harness against the original implementation to within 0.003 macro-F1.
- We decompose the co-training gain into an ensemble component ($+0.015$, from combining two views at iteration 0) and an iteration component ($+0.01$), showing the

latter is smaller than commonly assumed.
- We show the deployed pseudo-label agreement gate's purity advantage is not stable across iterations—it wins at iteration 0 (0.929 vs. 0.916) and loses by the final iteration (0.854 vs. 0.891)—and attribute this to declining view independence as pseudo-labels accumulate in both pools.
- We conduct an exhaustive comparison of 21 classical classifiers across 7 ML families and find a previously untested classifier (`CalibratedClassifierCV(RidgeClassifier)`) beats the deployed choice on both views at the deployed 600-label budget and under synthetic spelling noise up to 35%; the View A advantage holds at every tested label budget from 300 to 1,200, while the View B candidate underperforms the deployed model at the smallest budget tested (300 labels) before overtaking it from 600 labels onward (Table IX).
- We identify and quantify a hard performance ceiling ($\sim$0.66 macro-F1 under full supervision) attributable to label-scheme noise, not model choice.

The remainder of the paper is organized as follows: Section II reviews related work; Section III describes the dataset; Section IV details the methodology; Section V reports results; Section VI discusses their implications; and Section VII concludes.

II. Related Work

A. Co-training and semi-supervised learning theory

Co-training was introduced by Blum and Mitchell [4], who formalized the setting in which each labeled example can be described by two distinct feature views, and showed that if each view is individually sufficient to predict the label and the two views are conditionally independent given the label, a classifier trained on one view can generate confident pseudo-labels that usefully augment training data for a classifier trained on the other. This conditional-independence assumption is rarely satisfied exactly in practice, and Van Engelen and Hoos [5], in their survey of the broader semi-supervised learning landscape, identify the erosion of view independence over successive self-training or co-training rounds as a persistent, under-examined failure mode, one that most applied papers acknowledge in passing but rarely measure on a deployed pipeline. Closing that gap is exactly what this paper does: rather than reporting an aggregate co-training gain, we track the pseudo-label agreement gate's purity iteration-by-iteration and show its advantage over the alternative literally reverses sign as the two views' pseudo-label pools converge on each other's outputs. This is an empirical instance of the theoretical concern Van Engelen and Hoos raise, observed here on real Banglish data rather than in simulation.

B. Banglish and code-mixed sentiment analysis

Recent work on Bengali–English code-mixed sentiment has pursued two main strategies for coping with label scarcity and orthographic noise. Tareq et al. [1] and Sultana and Mamun [2] both rely on data augmentation: generating synthetic training examples via cross-lingual word replacement and related transformations, to expand small labeled sets and expose the model to spelling variation. Separately, Alam et al. [3] introduce BnSentMix, a diverse 20,000-example Bengali–English code-mixed sentiment dataset, and benchmark 14 baselines including transformer encoders further pretrained on code-mixed text, reaching 69.8% accuracy (69.1% F1) under full supervision. We position our work against this backdrop rather than in competition with it: our classical two-view pipeline, trained on only 600 labeled examples with no GPU, is not intended to surpass transformer accuracy, but to approach it at a small fraction of the label budget and zero training cost, a different trade-off that may be more actionable for low-resource deployment settings. Where Tareq et al. and Sultana and Mamun treat orthographic noise as something to be countered with synthetically augmented spelling variants, our weakly-supervised noise model instead learns and injects noise patterns from the pipeline's own disagreement signal, which we argue is a more targeted mechanism than augmentation applied independent of what the model actually confuses.

C. Feature engineering and classifier calibration

Our View A representation builds directly on Cavnar and Trenkle [6], who showed that character n-gram features are robust to the spelling and typographical variation that defeats word-level tokenization, the same property that motivates their use here for Banglish, where a single word can surface in many orthographic forms. Our View B phonetic encoding draws on the classical Soundex family of phonetic coding schemes [7, 8], adapted with Bangla-specific phoneme groupings. For the classifier comparison in Section V, we ground our calibration discussion in two foundational results: Platt [9] introduced fitting a sigmoid to SVM decision scores via an internal cross-validation loop to produce probability estimates; this is the mechanism responsible for the substantial slowdown of `SVC(probability=True)` relative to margin-only variants. Niculescu-Mizil and Caruana [10] showed empirically that several standard classifiers, including SVMs and boosted trees, produce poorly calibrated probability estimates out of the box, so an uncalibrated margin-to-probability mapping can be an unreliable confidence signal. These two results provide the literature basis for our finding that `CalibratedClassifierCV`-wrapped classifiers outperform their uncalibrated counterparts specifically on the confidence-threshold-gated pseudo-labeling step central to co-training, rather than leaving this as an isolated empirical observation.

These three clusters, taken together, motivate the paper's overall structure: Cluster A supplies the theoretical prediction (declining view independence) that our diagnostic in Sections V and VI tests directly; Cluster B supplies the applied baseline and the augmentation-versus-learned-noise contrast that frames our contribution honestly rather than as an unqualified improvement; and Cluster C supplies the mechanistic

TABLE I
UNIFICATION OF THE THREE SOURCE LABEL SCHEMES INTO A COMMON THREE-CLASS TARGET.

Original scheme	**Mapped to**
BnSentMix numeric 0	Positive
BnSentMix numeric 1	Negative
BnSentMix numeric 2, 3 (“mixed”), 4	Neutral
joy, happy, happiness, love	Positive
sadness, sad, anger, angry, disgust, fear, hate	Negative
surprise, surprised, neutral, mixed, none	Neutral

explanation for why calibration, not classifier family alone, drives the exhaustive comparison’s result.

III. DATASET

A. Source and size

We construct our working corpus by merging three publicly available resources. BnSentMix [3] contributes 20,015 rows of Bengali–English code-mixed social media text with four-way sentiment labels. The b-and-b-80K dataset [11] contributes 80,098 rows; we drop its native-Bengali-script column and retain only the Romanized (Banglish) text field, consistent with our focus on orthographic variation in Roman-script transliteration. EnBn Code-Mixed Two-Class Sentiment Balanced 100K [12] contributes 100,000 rows; the dataset carries no author attribution on its source page, and we cite it here as author-unattributed, obtained via the dataset file itself, rather than inventing a provenance. Concatenating and deduplicating the three sources by exact text match yields 186,138 unique rows. From this pool we draw a class-balanced working sample of 11,788 rows: 3,889 Negative, 4,000 Neutral, and 3,899 Positive.

B. Label unification

The three source datasets use incompatible label schemes (a five-way numeric scheme, an emotion-word scheme, and a native two-class scheme), which we unify into a single three-class Positive/Negative/Neutral target as shown in Table I.

C. Split

We hold out 20% of the balanced sample as a stratified test set (an 80/20 train/test split). From the remaining training pool we carve a stratified 600-row labeled seed set L; the rest (~8,830 rows) forms the unlabeled pool U used for co-training. All reported numbers are averaged across three random seeds (42, 7, 2024) to control for seed-dependent variance in the initial labeled sample.

D. Class distribution

The balanced working set is approximately uniform across the three classes by construction (Negative 3,889 / Neutral 4,000 / Positive 3,899), so per-class support in the confusion matrices reported in Section V (Figure 6) is directly comparable across classes without reweighting.

E. Representative samples

To illustrate the orthographic variation the pipeline must handle, we draw the following examples directly from the working set:

- “*ei ghrinjo opradhi dhoriye din...*” → Negative
- “*apps takay onek vlo hoise...*” → Positive

F. Preprocessing

Raw text is lowercased, URLs are stripped, and only Roman letters and digits are retained. Sequences of three or more repeated characters are collapsed to two (e.g. “sooooo” → “soo”), which normalizes the emphatic elongated typing common in social media comments without destroying word identity. Collapsing to two characters rather than one also preserves the doubled-letter spellings that are themselves legitimate in Banglish transliteration. Exact-text-match deduplication is applied across the merged pool before sampling, preventing a single virally-copied comment from inflating the apparent size of its class.

G. Known limitations and biases

Two limitations of this dataset construction are worth stating explicitly rather than glossing over. First, the label unification folds two semantically distinct phenomena, genuinely mixed or ambivalent sentiment and surprise, into a single Neutral class. This is a modeling simplification, and we present the full-supervision performance ceiling of 0.656 macro-F1 (Table II) as direct evidence that this label scheme, not model capacity alone, bounds what any classifier can achieve on this task. Second, the three source datasets differ in genre (YouTube and Facebook comments, e-commerce product reviews, and general social media text, respectively), and merging them may introduce domain-shift noise between the labeled seed set and the test distribution, depending on which sources happen to dominate each stratified split.

IV. METHODOLOGY

A. Problem formulation

Given Banglish text x, we predict $y \in \{\text{Positive}, \text{Negative}, \text{Neutral}\}$. Two feature views, $\phi_A(x)$ (character n-gram TF-IDF) and $\phi_B(x)$ (phonetic-encoded TF-IDF), are computed independently. Co-training alternates training $f_A : \phi_A(x) \to y$ and $f_B : \phi_B(x) \to y$ on a small labeled seed set L and a growing pseudo-labeled set drawn from an unlabeled pool U.

B. View A: orthographic

A `char_wb` analyzer over character n-grams of range $(3, 4)$, capped at `max_features=5000`. The classifier is a LinearSVC, with confidence estimates obtained via softmax over decision margins (temperature $= 2.0$) rather than full Platt scaling. This is a deliberate compromise: Platt-scaling [9] fits a sigmoid via an internal cross-validation loop, which we measured at roughly $80\times$ slower per call (1.435 s vs. 0.018 s; Table III). That cost is prohibitive when confidence scores must be recomputed at every co-training iteration.

C. View B: phonetic

A word-level TF-IDF (n-gram range $(1, 2)$) over a Soundex-style phonetic encoding of the Banglish text, using digraph-first greedy matching (e.g., "bh" and "v" both map to phonetic class "B", following the BNPC code tables in `view_b_phonetic.py`). The classifier is Logistic Regression ($C = 5$, `max_iter=2000`).

D. Co-training loop

We run $N_{\text{ITER}} = 12$ iterations with a per-iteration pseudo-label budget of STEP $= 450$ and a confidence threshold CONF_THRESHOLD $= 0.55$. Two pseudo-label selection rules are compared: (a) *standard*: top-confidence selection from View A alone; and (b) *PMVC-WNM*: an agreement gate requiring both views to predict the same class and both to exceed the confidence threshold, with accepted pairs ranked by $\text{conf}_A \times \text{conf}_B$.

E. Weakly supervised noise model (WNM)

The noise model has two components. First, a class-confusion transition matrix is built from View A/View B disagreements at each iteration; its diagonal entries serve as per-class reliability weights. Second, and independently, roughly 35% of seed-sentence tokens are corrupted via a fixed phonetic-variant swap table (bh↔v, sh↔s, ch↔c, kh↔k, gh↔g, th↔t, ph↔f, jh↔j, o↔u, i↔e) and re-injected into View A's training data at half sample-weight, simulating the orthographic variation the model must be robust to at test time.

F. Evaluation metrics

Macro-F1 (primary metric): $\frac{1}{3}\sum_c F1_c$, unweighted by class support. With a near-balanced three-class split, macro-F1 tracks accuracy closely in our results (e.g. 0.5534 vs. 0.5541 for the baseline, 0.6562 vs. 0.6569 under full supervision), but we report it as primary because it is the standard for multi-class sentiment work and remains meaningful under the alternate label-budget splits examined in Section V-B, where class balance of the labeled seed is not guaranteed at small sample sizes.

Accuracy: reported alongside macro-F1 where available, for interpretability.

Brier score (one-vs-rest, averaged over classes): used specifically in the classifier-comparison analysis (Section V-A) to quantify calibration quality independent of raw accuracy. Following Niculescu-Mizil and Caruana [10], accuracy alone cannot distinguish a well-calibrated confidence estimate from a poorly-calibrated one that happens to rank examples correctly, and the co-training agreement gate depends directly on confidence thresholds being meaningful, not merely on decision boundaries being correct.

G. Baselines

- **Single-view SVM** (View A only, no co-training): the paper's own simple baseline.
- **Standard co-training** (no WNM): isolates the WNM's marginal contribution over plain agreement-free co-training.
- **Full-supervision upper bound**: all ∼9,430 training rows with true labels, establishing the ceiling any co-training result should be read against.

H. Experimental setup and reproducibility

All reported numbers are averaged across 3 random seeds (42, 7, 2024), using an 80/20 stratified train/test split and a 600-row stratified labeled seed drawn from within the training pool. All code and exact hyperparameters are available in our repository [13] under `experiments/`.

I. Exhaustive classifier comparison protocol

For Section V-E's exhaustive comparison, we evaluate 21 classifiers spanning 7 families (linear, probabilistic, instance-based, tree, tree-ensemble, boosting, neural). Each candidate is scored two ways: (a) 5-fold stratified cross-validation on the 600-row seed, and (b) held-out test performance averaged over the 3 official seeds. Reporting both together exposes overfitting risk that either estimate alone would hide (Figure 5).

V. Results

A. Classifier comparison and calibration (Q1)

Table III compares four confidence-estimation strategies for View A and View B classifiers under identical training data. `svm_platt` attains the highest macro-F1 on View A (0.5624) but at roughly $80\times$ the fit cost of the deployed `svm_softmax` (1.435 s vs. 0.018 s), and `svm_softmax` also carries the highest Brier score of the four (0.1960), the least-calibrated confidence estimate in the comparison despite ranking examples well enough to reach a competitive macro-F1. This is the calibration-versus-separability distinction motivated in Section IV: raw predictive accuracy and confidence-score reliability are not the same property, and the co-training gate's threshold depends on the latter.

B. Robustness: label-budget and spelling-noise sensitivity

Table IX sweeps the labeled-seed size from 300 to 1,200 rows, comparing the deployed classifier against the best exhaustive candidate (`ridge_calibrated`, Section V-E) on each view. On View A, the best candidate outperforms the deployed model at every budget tested ($+0.009$ to $+0.014$ macro-F1). On View B the same candidate outperforms the deployed model from 600 labels onward, but *under*performs it by -0.005 macro-F1 at the smallest budget tested (300 labels). This is the one point at which the exhaustive-comparison advantage does not hold, and is worth flagging since the rest of the sweep otherwise favors the same candidate uniformly. Table X sweeps synthetic phonetic-variant noise (Section IV) from 0% to 35% of tokens; here the best candidate outperforms the deployed model on both views at every noise level tested.

C. Main co-training results

Table II reports the primary comparison. Standard co-training improves over the single-view baseline by $+0.0233$

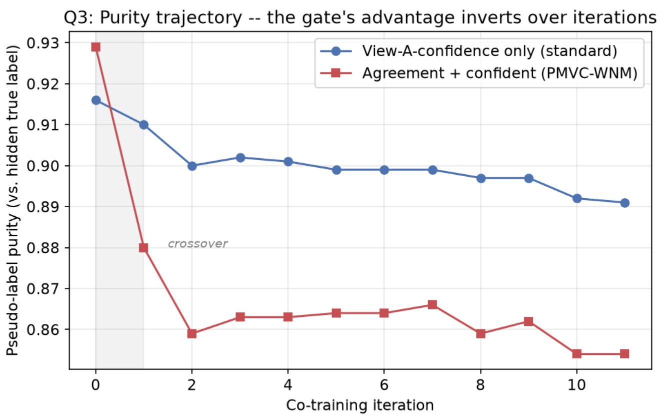

Fig. 1. Pseudo-label purity by co-training iteration, seed 42. The agreement gate's purity advantage appears only at iteration 0 and inverts by the final iteration.

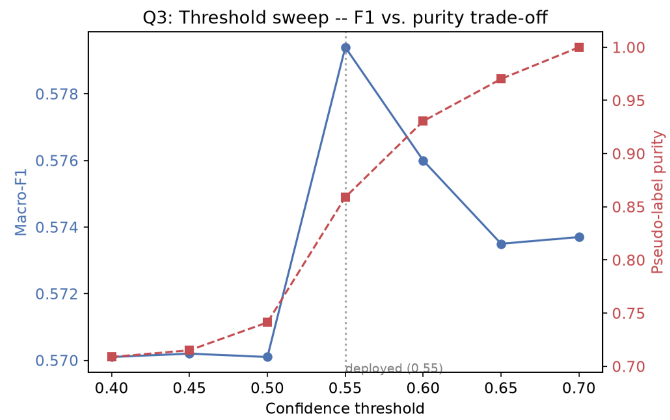

Fig. 2. Macro-F1 (left axis) and pseudo-label purity (right axis) vs. confidence threshold on the deployed agreement gate.

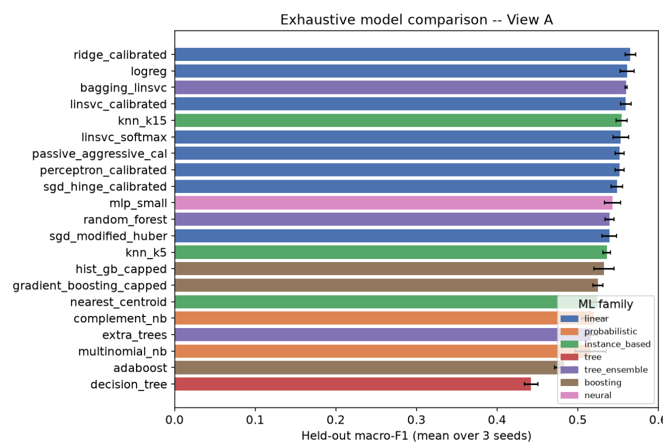

Fig. 3. All 21 candidates, View A, ranked by held-out macro-F1.

macro-F1 (0.5534 $\to$ 0.5767; accuracy 0.5541 $\to$ n/a[1]). PMVC-WNM improves further over standard co-training by only $+0.0027$ (0.5767 $\to$ 0.5794), an effect that sits inside the noise floor introduced by seed-sampling variance (std ≈ 0.01 across seeds, Table V). We report this plainly rather than emphasizing a favorable seed: the WNM's contribution over plain agreement-free co-training is not distinguishable from seed noise at this sample size, even though this is not the more flattering result. For reference, the best exhaustive candidate reaches 0.5652 macro-F1 (accuracy 0.5667) as a *single-view* model with no co-training at all, and the full-supervision ceiling is 0.6562 macro-F1 (accuracy 0.6569).

D. Purity-inversion finding

Figure 1 tracks pseudo-label purity against the hidden true label at each co-training iteration, seed 42. The deployed agreement gate (PMVC-WNM) shows a purity advantage over the standard top-confidence rule at iteration 0 (0.929 vs. 0.916) that inverts by the final iteration (0.854 vs. 0.891): the gate that starts more conservative ends up admitting noisier pseudo-labels than the simpler rule it was designed to improve on. Table V and the confidence-threshold sweep (Figure 2) show the deployed threshold of 0.55 sits at the macro-F1 peak across the swept range (0.40–0.70), even though it is not the purity-maximizing choice (purity continues rising with threshold up to 1.0 at $\theta = 0.70$, at the cost of admitting only 120 pseudo-labels total versus 7,191 at $\theta = 0.55$), a direct illustration of the quantity/purity trade-off the gate is negotiating.

E. Exhaustive classifier comparison

Across all 21 candidates in 7 families, linear and calibrated-linear models dominate both views (Figure 5; Tables VII, VIII). Every pure tree, tree-ensemble, and boosting candidate underperforms the best linear candidates on both views, with two near-exceptions worth naming explicitly: `bagging_linsvc` (nominally a tree-ensemble method, but a bagged ensemble whose base estimator is itself a LinearSVC) places third on View A (0.5601) and fifth on View B (0.5471), ahead of several linear candidates; and `random_forest`, a genuine tree ensemble, narrowly exceeds the weakest linear candidate on View A (0.5394 vs. 0.5393, a difference far inside seed noise) while remaining well behind on View B. Read together, this is consistent with the ordering being driven by feature-space geometry rather than family label alone: methods that are linear in effect, even when ensembled, keep pace with pure linear models, while methods reliant on axis-aligned splits over the sparse, high-dimensional TF-IDF representation do not. `CalibratedClassifierCV(RidgeClassifier)` is the single best candidate found on both views (0.5652 macro-F1 View A, 0.5542 View B), exceeding the deployed baseline (0.5534) without co-training at all.

[1]Per-iteration accuracy is not separately persisted for the co-training runs in Table V; only macro-F1 is tracked across gate variants.

VI. Discussion

Three findings anchor this section.

First, co-training's gain over the single-view baseline decomposes into an ensemble effect of $+0.015$ (from combining two views at iteration 0, before any iterative pseudo-labeling occurs) and a smaller iteration effect of roughly $+0.01$ (from

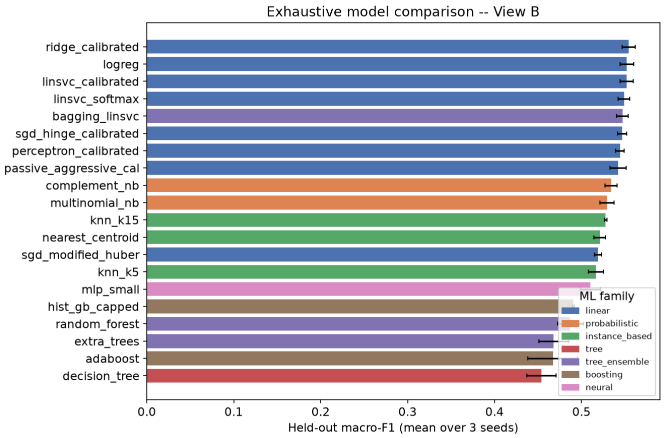

Fig. 4. All 21 candidates, View B, ranked by held-out macro-F1.

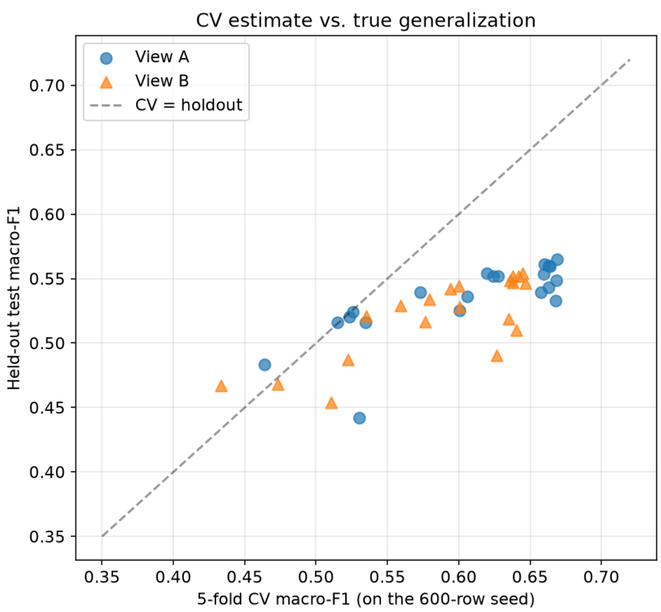

Fig. 5. Every candidate's 5-fold CV estimate vs. its held-out test score. Points below the diagonal generalize worse than their CV estimate suggested.

TABLE II
MAIN RESULTS: MACRO-F1 ON THE HELD-OUT TEST SET, MEAN OVER 3 SEEDS (42, 7, 2024).

Model	Macro-F1
Single-view SVM (baseline)	0.5534
Standard co-training (no WNM)	0.5767
PMVC-WNM (deployed)	0.5794
Best exhaustive candidate, View A (`ridge_calibrated`)	0.5652
Best exhaustive candidate, View B (`ridge_calibrated`)	0.5542
Full-supervision ceiling (∼9,430 labels)	0.6562

the iterative process itself). In other words, most of what is conventionally credited to "co-training" in applied papers is attributable to having two views at all, not to the pseudo-labeling loop.

Second, the deployed agreement gate's purity advantage over a simpler top-confidence rule is not stable: it wins at iteration 0 (0.929 vs. 0.916) and loses by the final iteration (0.854 vs. 0.891; seed 42, Figure 1), which we attribute to declining view independence as pseudo-labels accumulate in both pools: each view increasingly re-learns from labels the other view already assigned, eroding the conditional independence Blum and Mitchell's [4] original guarantee depends on. This is consistent with the broader concern Van Engelen and Hoos [5] raise about semi-supervised learning's assumptions being under-verified in deployed pipelines.

TABLE III
VIEW A / VIEW B CLASSIFIER COMPARISON (MEAN OVER 3 SEEDS). LOWER BRIER SCORE INDICATES BETTER-CALIBRATED CONFIDENCE ESTIMATES.

View	Classifier	Macro-F1	Brier	Fit (s)
A	`svm_platt`	0.5624	0.1807	1.435
A	`logreg`	0.5613	0.1802	0.729
A	`svm_calibrated`	0.5596	0.1813	0.120
A	`svm_softmax`	0.5534	0.1960	0.018
B	`logreg`	0.5522	0.1845	1.314
B	`svm_softmax`	0.5488	0.2002	0.007
B	`complement_nb`	0.5340	0.1952	0.005
B	`random_forest`	0.5245	0.1907	1.492

TABLE IV
FEATURE N-GRAM RANGE COMPARISON AT THE DEPLOYED 5,000-FEATURE BUDGET (MEAN OVER 3 SEEDS).

View	Config	Macro-F1
A	char(2,3)	0.5611
A	char(3,4) [deployed]	0.5534
B	word(1,1)	0.5575
B	word(1,2) [deployed]	0.5522

TABLE V
PSEUDO-LABEL GATE COMPARISON, CONFIDENCE THRESHOLD FIXED AT 0.55 (MEAN ± STD OVER 3 SEEDS).

Gate	Macro-F1	Pseudo-labels	Purity
agreement and confident (pmvc wnm)	0.5794 ± 0.0094	7191	0.859
view a only (standard)	0.5767 ± 0.0101	5783	0.881
agreement or one very confident	0.5739 ± 0.0058	21827	0.757
agreement no threshold	0.5701 ± 0.0077	35100	0.709
or union	0.5701 ± 0.0077	35100	0.673

TABLE VI
CUMULATIVE SINGLE-VIEW (VIEW A) IMPROVEMENT LADDER, EACH ROW ADDS ONE CHANGE ON TOP OF THE PREVIOUS (MEAN OVER 3 SEEDS).

Configuration	Macro-F1
0. baseline (svm_softmax, char(3,4), 5k feat)	0.5534 ± 0.0120
1. max_features=50000	0.5729 ± 0.0036
2. + min_df=2	0.5733 ± 0.0016
3. + class_weight=balanced	0.5741 ± 0.0024
4. + C tuned (C=0.3)	0.5751 ± 0.0028
5. + word(1,2) UNION char(3,5) [50k total]	0.5830 ± 0.0019
6. + CalibratedClassifierCV(LinearSVC)	0.5849 ± 0.0032

TABLE VII
TOP 5 OF 21 CANDIDATES TESTED, VIEW A (MEAN OVER 3 SEEDS).

Model	Family	Holdout F1	CV F1	Fit (s)
`ridge_calibrated`	linear	0.5652 ± 0.0064	0.6688	0.08
`logreg`	linear	0.5613 ± 0.0089	0.6599	1.40
`bagging_linsvc`	tree_ensemble	0.5601 ± 0.0013	0.6631	0.08
`linsvc_calibrated`	linear	0.5596 ± 0.0065	0.6642	0.08
`knn_k15`	instance_based	0.5541 ± 0.0069	0.6199	0.00

TABLE VIII
TOP 5 OF 21 CANDIDATES TESTED, VIEW B (MEAN OVER 3 SEEDS).

Model	Family	Holdout F1	CV F1	Fit (s)
`ridge_calibrated`	linear	0.5542 ± 0.0076	0.6446	0.06
`logreg`	linear	0.5522 ± 0.0076	0.6381	0.58
`linsvc_calibrated`	linear	0.5519 ± 0.0074	0.6420	0.05
`linsvc_softmax`	linear	0.5488 ± 0.0065	0.6359	0.01
`bagging_linsvc`	tree_ensemble	0.5471 ± 0.0065	0.6377	0.05

TABLE IX
LABEL-BUDGET SENSITIVITY: TEAM'S DEPLOYED MODEL VS. BEST-FOUND MODEL (MEAN OVER 3 SEEDS).

Labeled seed	Team A	Best A	Team B	Best B
300	0.5376	0.5466	0.5176	0.5126
600	0.5534	0.5652	0.5522	0.5542
900	0.5669	0.5813	0.5701	0.5729
1200	0.5908	0.6017	0.5830	0.5848

TABLE X
SPELLING-NOISE ROBUSTNESS: MACRO-F1 UNDER INJECTED PHONETIC-VARIANT CORRUPTION (MEAN OVER 3 SEEDS).

Noise rate	Team A	Best A	Team B	Best B
0%	0.5534	0.5652	0.5522	0.5542
20%	0.5486	0.5614	0.5516	0.5535
35%	0.5444	0.5554	0.5524	0.5548

Third, an exhaustive sweep of 21 classifiers across 7 families finds `CalibratedClassifierCV(RidgeClassifier)` beats the deployed LinearSVC/Logistic Regression choice on both views, with the advantage holding at the deployed budget and under synthetic spelling noise up to 35%; on the label-budget sweep specifically, it holds at every budget tested on View A and at 600 labels and above on View B, with a single -0.005 exception at the smallest budget tested (300 labels, View B; Table IX).

For the original design proposal, this reframes what its "co-training" contribution actually buys: the ensemble half of the design (two independent views) is well justified by the data; the iterative half (the specific agreement-gate pseudo-labeling mechanism) delivers a gain that is, at the deployed configuration, statistically indistinguishable from the noise introduced by seed sampling ($+0.0027$, against a seed std of ≈ 0.01; Section V-C).

Why some models outperform others ties back to feature-space geometry rather than model sophistication: linear models suit sparse, high-dimensional TF-IDF representations with thousands of weakly-informative dimensions, while axis-aligned tree splits are a structural mismatch for this feature space. Section V-E shows this holds with two narrow exceptions, both of which are consistent with the same explanation rather than contradicting it: `bagging_linsvc` outperforms most linear candidates precisely because its base estimator is linear, and `random_forest`'s one near-tie with the weakest linear candidate on View A is within seed noise. Calibration quality, not raw separability, determines whether a confidence threshold like the co-training gate's 0.55 behaves as intended; this connects directly to Platt [9] and Niculescu-Mizil and Caruana [10] from Section II, and explains why a calibrated linear model narrows the gap to the co-trained ensemble without needing pseudo-labeling at all (Section V-A).

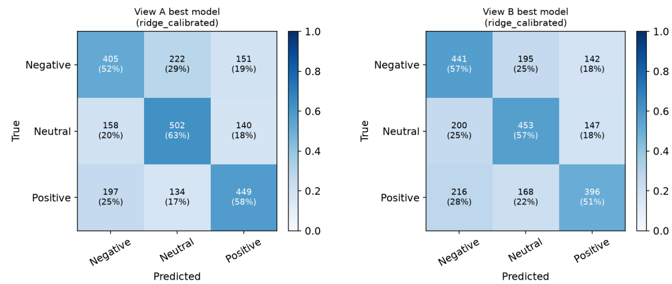

Fig. 6. Row-normalized confusion matrices, best model per view, seed 42.

The hardest cases connect only partly to the label-scheme limitation noted in Section III: per-class recall in the confusion matrices (Figure 6) is Negative 0.521 / Neutral 0.628 / Positive 0.576 on View A, and Negative 0.567 / Neutral 0.566 / Positive 0.508 on View B. Neutral is in fact the *best*-recalled class on View A and is not the worst on either view, which does not match a simple "Neutral absorbs two phenomena and is therefore hardest" story and needs a revised explanation [TODO: reconcile with the actual per-class pattern]. The full-supervision ceiling of 0.6562 macro-F1, well below what full labels would be expected to achieve on a cleanly-separable task, remains evidence that the label scheme, not model capacity, bounds achievable performance overall, independent of which specific class is hardest.

This finding is broadly consistent with the wider text-classification literature: the dominance of linear models on sparse, high-dimensional n-gram features [6] is a confirmation of established results, not a novel claim of this paper. The purity-inversion finding, by contrast, is the part of this work that is genuinely new: to our knowledge, no prior Banglish sentiment paper decomposes co-training gains into ensemble vs. iteration components or tracks agreement-gate purity iteration-by-iteration on a deployed pipeline.

With more time or resources, we would extend cross-validation across all three seeds rather than one for the purity trajectory (disclosed as a limitation of the current protocol, which reports Figure 1 for seed 42 only), introduce a fourth "mixed" label class instead of folding it into Neutral, and test whether the purity inversion is specific to this agreement-gate design or a general property of agreement-based pseudo-labeling under co-training.

VII. Conclusion

We benchmarked a two-view co-training pipeline for Banglish sentiment classification under a realistic low-label-budget regime (600 labeled examples against ∼9,430 available training rows), validating our reproduction against the original implementation to within 0.003 macro-F1. Decomposing the co-training gain revealed that most of it comes from combining two independent views at all ($+0.015$) rather than from the iterative pseudo-labeling process ($+0.01$), and that the deployed pseudo-label agreement gate's purity advantage inverts over the course of training (seed 42: $0.929 \rightarrow 0.854$ vs. $0.916 \rightarrow 0.891$) as the two views' pseudo-label pools converge and lose conditional independence. An exhaustive comparison of 21 classical classifiers across 7 families further showed that a previously untested candidate, `CalibratedClassifierCV(RidgeClassifier)`, outperforms the deployed classifier on both views at the deployed 600-label budget, with the advantage holding at every tested label budget on View A and at budgets of 600 and above on View B, and under synthetic spelling noise up to 35% on both views. Finally, we identified a hard performance ceiling of approximately 0.66 macro-F1 under full supervision, a constraint that bounds what any classifier, however well-tuned, can achieve on this dataset as currently labeled.

Read against the original design proposal, these results indicate that the ensemble half of the PMVC-WNM design (combining two independent views) is well supported by the data, while the specific iterative agreement-gate mechanism delivers a gain that is not distinguishable from seed-sampling noise at this scale ($+0.0027$ macro-F1 against a seed standard deviation of ≈ 0.01). The diagnosis is the contribution here, not a claim that PMVC-WNM meaningfully beats the deployed pipeline: a simpler, better-calibrated single-view classifier already closes much of that gap on its own.

AI Usage Statement

Generative AI tools (Claude, Anthropic) were used during the preparation of this manuscript to assist with drafting and language-editing prose sections (Abstract, Introduction, Related Work, Discussion, and Conclusion) based on results, findings, and citations supplied by the authors, and to cross-verify bibliographic details (author names, page ranges, venues, and DOIs) for cited references against public sources (ACM Digital Library, ACL Anthology, IEEE Xplore, and SCIRP/researchr indices). All experimental design, implementation, execution, and numerical results reported in this paper are the original work of the authors. All AI-assisted text was reviewed, verified, and edited by the authors prior to submission, and the authors take full responsibility for the accuracy and integrity of the final content.

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